



Instructor: Date:

Learner: Regional Branch:

Program: Center:

CONTENT

- Let's Start! • Let's Explore! • Let's Practice! • Extension Activity • Target Words • Learning Points



Let's Start!

Types of Software Testing and Its Functionality



Learning Outcome:

Implement and document the software tests, applying trial and error, based on the designed plan and procedures established by the company.

Applies the different types of software testing to the IT solution.

Ask your partner:

- Do you think it's important to run different software tests regardless the software system? Why?
- What are the advantages and disadvantages of software testing?

Can Do Statements:

- I can talk about unit testing, integration testing and functional testing.
- I can describe system testing and sanity testing.
- I can explain what regression testing, load testing and usability testing are.





Let's Explore!

Listen to the explanation about different types of software testing.



Instructions: Read the text below.

If you want to run the right test on your software, you will have to make sure you know its characteristics and possible flaws. Though a software system may undergo different testing before approval, it is important not to waste time performing unnecessary tests. Different types of software testing are explained in order of how much they encompass the system to be tested.

If you want to perform white box testing, you will have to go for unit testing first, as it checks individual software components. It is usually done by the programmer since it requires deep knowledge of software and codes. The following testing types belong to the black box testing.

Integration Testing: Testing of a group of components that work together to produce the desired output. Hardware and software are tested if they are linked to each other.

Functional Testing: It focuses on output; though it, you can check if the system fulfills the requirements.

Regression Testing: It checks the system after the modification is done to see if the corrections run well and don't hinder another process.

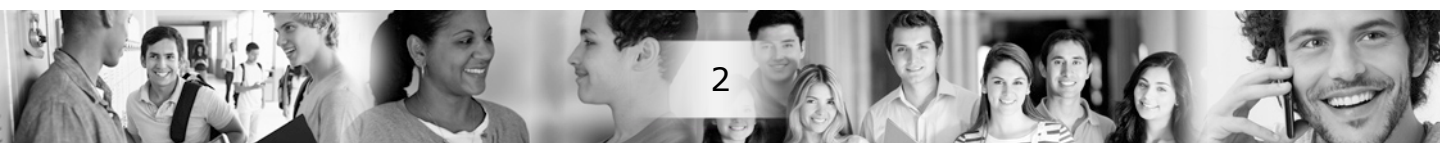
Sanity Testing: It belongs to regression testing, and it checks the functionality of the system after correction has happened and bugs have been fixed properly.

Load Testing: It assesses system performance under heavy loads and how it affects the time response in the system.

System Testing: Software is evaluated entirely to see if it works well in different environments.

Usability Testing: It checks how friendly the system is to different users.

Remember that if you choose the most suitable software test, you will save tons of money and human resources. The tests listed here are just some of the many others you can find and apply to different kinds of programs.





Let's Practice!

System and sanity testing are great options to check functionality.

Instructions: organize the mystery sentence.

1. know / wanted / you / you / you / would / If / flaws. / attributes / have / the /
make / possible / to / sure / to / its / right / and / run / test,

2. sanity / use / If / the / to / the / had / testing. / the / system / you / correction /
the / to / you / functionality / happened, / needed / would've / before / of / check /
intended

3. to / software / environments, / a / do / have / you'll / in / If / different / you / to /
want / if / well / testing. / works / system / the / see



Let's Practice!

Let's go over* the next testing types which belong to the black box testing.

* To "go over"* means to analyze and check a topic.

Instructions: Select the correct answer

1. The system _____ through the regression testing.
a. is checked b. are checked c. checks d. checked
2. The regression testing takes place after modification _____ to see if the
corrections run well and don't hinder another process.
a. are done b. is done c. done d. are made
3. The system performance _____ by the load testing; it assesses heavy loads
and how they affect the time response in the system.
a. is assessed b. assessed c. are assessed d. to be assessed
4. The usability testing _____ to check how friendly the system is to different users.
a. are used b. were used c. is use d. is used
5. White box testing checks individual software components and is usually _____
the programmer.
a. made by b. make by c. done by d. do by





Extension Activity

It's time for you to show how much you've learnt about software testing.

Instructions: Complete the following steps. Fill in the Self-Assessment.

- Look for more information about the types of software testing you studied throughout this lesson.
- For each testing method, write down a couple of disadvantages you found about them.
- Create a chart in which you present the disadvantages and include a possible solution for each. Don't forget to use the grammar practiced in the lesson.
- Discuss the chart with a partner.

Self-Assessment

Yes No

- I look for more information about the types of software testing I studied throughout this lesson. ☐ ☐
- For each testing method, I write down a couple of disadvantages I found about them. ☐ ☐
- I create a chart in which I present the disadvantages and include a possible solution for each. I use the grammar practiced in the lesson. ☐ ☐
- I discuss the chart with a partner. ☐ ☐



Target Words

- **flaw** (noun): a defect or an imperfection.
e.g. If you want to run the right test on your software, you will have to make sure you know its characteristics and possible flaws.
- **undergo** (verb): to experience or go through something.
e.g. Though a software system may undergo different testing before approval, it is important not to waste time performing unnecessary tests.
- **encompass** (verb): to cover or to include.
e.g. Different types of software testing are explained in order of how much they encompass the system to be tested.
- **fulfill** (verb): to meet the expectations.
e.g. It also focuses on output; though it, you can check if the system fulfills the requirements.
- **hinder** (verb): to represent an obstacle or make something difficult to achieve.
e.g. It checks the system after modification is done to see if the corrections run well and don't hinder another process.
- **heavy load** (noun): in systems, it means that there are many processes happening at the same time.
e.g. It assesses system performance under heavy loads and how it affects the time response in the system.



Learning Points

Conditional Sentences with If Clauses- Affirmative Form

Conditional sentences state a relation between something that can take place if certain conditions are/were met.

Conditional clauses talk about situations that need a condition to be true. The final result depends on the condition to be real.

Conditional sentences are made of 2 clauses: the **"if clause"** that explains the condition, and the main clause that presents the result or consequence based on the **"if clause"**.

Any of the two clauses can be affirmative, negative or even mixed. You just have to be clear about the meaning and the idea you want to express.

Conditional type 1

In order to make an affirmative statement using first conditional, use the following structure:

1. **If + present simple + , + future simple / modals can, may, might, must**
2. **future simple / modals can, may, might, must + If + present simple**

Examples:

*If you **want to** implement software in the company, it **will** have to be tested.*

*Software **will** have to be tested first **if you want to** implement it in the company.*

*If you **want to** implement software in the company, it **must be** tested first.*

*Software **must** be tested first **if you want to** implement it in the company.*

Conditional type 2

The conditional type 2 or second conditional is used for expressing unreal situations that may take place in the present.

Structure:

If + past simple + , + would/could.
would/could + If + past simple.

Examples:

*If I **tried to** implement a testing system over this software without letting my boss know, he **would** kill me immediately.*

*My boss **would** kill me immediately if I **tried to** implement a testing system over this software without letting him know.*

Notice that even though the main verb of the if clause is used in past form, the meaning is given to the unreal situation at the present.





Learning Points

Conditional Type 3

Make use of this type of conditional to express unreal situations that didn't happen at a past time.

Structure:

If + past perfect + , + would/could + present perfect.

would/could + present perfect + If + past perfect.

Examples:

If I had known that implanting a black box testing was that easy, I would've finished testing the app before that day.

I would've finished testing the app before that day if I had known that black box testing was that easy to implement.

Examples

- *If you want to run the right test on your software, you will have to make sure you know its characteristics and possible flaws.*
- *If you want to perform white box testing, you will have to go for unit testing first.*
- *Remember that if you choose the most suitable software test, you will save tons of money and human resources.*

Unit Testing, Integration Testing, and Functional Testing

There are many types of software systems that can be performed at different stages and in different software systems.

The following testing types describe what is to be tested in a software system that undergoes evaluation.

The first one is **the unit testing**, that evaluates an individual unit or a group of units where the given input is contrasted against the expected output.

The second is **the integration testing**, which stands for a group of components within a software system that works together to produce a common output. Sometimes hardware and software may be linked to each other, and in that case hardware is tested as well. This kind of test is done to corroborate that software works well after the integration of the different components has taken place.

The last one is **the functional testing**, and as its name says so, it focuses on whether the required functionality works or not. Different tests may be run to different functions.